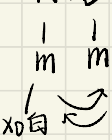
第二次作业 习题三 1、3、5、7-8

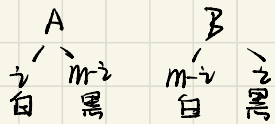
1、 A、B 各 m 白 m 黑



(1) 时齐马尔可夫链：未来与过去无关，只与现在有关，且 P_{ij} 不依赖于 n

一步转移概率：

$$P_{ij}=\begin{cases}\frac{i}{m}\cdot\frac{i}{m}, & j=i-1 \quad \text{白球减少，黑球增加}\\ \frac{i}{m}\cdot\frac{m-i}{m}\times 2, & j=i \quad \text{白球不变}\\ \frac{m-i}{m}\cdot\frac{m-i}{m}, & j=i+1 \quad \text{白个黑}\downarrow\end{cases}$$



X_n 与 X_{n-2} X_{n-3} $\cdots$ X_0 无关，只与 X_{n-1} 有关

P_{ij} 不依赖于 n，所以 $\{X_n\}$ 为时齐马尔可夫链

状态空间：$I=\{0,1,2\cdots m\}$

3、累计连续为 1 的次数 L_n

状态空间：$I=\{0,1,2\cdots\}$

$$P_{ij}=\begin{cases}1-P, & j=0\\ P, & j=i+1\end{cases}$$

5、 (1) $P(Y_2=1\mid Y_0=1,Y_1=b)=\dfrac{P(Y_0=1,Y_1=b,Y_2=1)}{P(Y_0=1,Y_1=b)}=\dfrac{P\begin{pmatrix}\max\{X_1,X_2\}=1\\ \max\{X_2,X_3\}=b\\ \max\{X_3,X_4\}=1\end{pmatrix}}{P\begin{pmatrix}\max\{X_1,X_2\}=1\\ \max\{X_2,X_3\}=b\end{pmatrix}}$ ⇒ 不可能 $=0$

$$P(Y_2=1\mid Y_1=b)=\frac{P(Y_2=b,Y_1=b)}{P(Y_1=b)}=\frac{P(\max\{X_2,X_3\}=b\cdot\max\{X_3,X_4\}=1)}{P(\max\{X_1,X_2\}=b)}$$

$$=P(X_2=b,X_3=b,X_4=1)$$

$$= \frac{\frac{1}{6}\times\frac{1}{6}\times\frac{1}{6}}{P(X_2=6 \text{ 或 } X_3=6)}$$

$$= \frac{\frac{1}{6}\times\frac{1}{6}\times\frac{1}{6}}{\frac{1}{6}+\frac{1}{6}\times\frac{5}{6}} = \frac{\frac{1}{6}\times\frac{1}{6}}{1+\frac{5}{6}} = \frac{1}{36}\times\frac{6}{11} = \frac{1}{66}$$

(2) $P(Z_2=12 \mid Z_0=2, Z_1=7) = \frac{P(Z_2=12, Z_0=2, Z_1=7)}{P(Z_0=2, Z_1=7)} = \frac{P(X_3+X_4=12, X_1+X_2=2, X_2+X_3=7)}{P(X_1+X_2=2, X_2+X_3=7)}$

$= \frac{P(X_1=1, X_2=1, X_3=6, X_4=6)}{P(X_1=1, X_2=1, X_3=6)} = \frac{1}{6}$

$P(Z_2=12 \mid Z_1=7) = \frac{P(X_2+X_3=7, X_3+X_4=12)}{P(X_2+X_3=7)} = \frac{P(X_2=1, X_3=6, X_4=6)}{\frac{1}{6}\times\frac{1}{6}\times 6}$

1	6
2	5
3	4
4	3
5	2
6	1

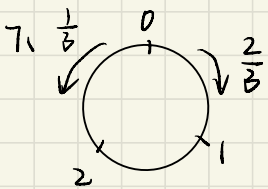
$= \frac{\frac{1}{6}\times\frac{1}{6}\times\frac{1}{6}}{\frac{1}{6}\times\frac{1}{6}\times 6}$

$= \frac{1}{36}$

(3) $P(Y_2=1 \mid Y_0=1, Y_1=6) \neq P(Y_2=1 \mid Y_1=6)$

$P(Z_2=12 \mid Z_0=2, Z_1=7) \neq P(Z_2=12 \mid Z_1=7)$

∴ $\{Y_n\}$ 和 $\{Z_n\}$ 都不具有马尔可夫性，都受过去影响.

7. 

(1) $P = \begin{pmatrix} 0 & \frac{2}{3} & \frac{1}{3} \\ \frac{1}{3} & 0 & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} & 0 \end{pmatrix}$

(2) $P(X_0=0, X_2=0, X_4=1) = \frac{1}{2}\cdot\frac{2}{3}\times\frac{2}{3}\cdot\frac{1}{3}\times\frac{1}{3}$

$= P(X_0=0)\, P_{00}^{(2)} \cdot P_{01}^{(2)}$ $= \frac{2}{81}$

$$P(X_2=1) = P_{01}^{(2)} + P_{11}^{(2)} + P_{21}^{(2)}$$

$$= \frac{1}{2}\cdot\left(\frac{1}{3}\right)^2 + \frac{1}{4}\cdot\frac{2}{3}\times\frac{1}{3}\times 2 + \frac{1}{3}\times\frac{1}{3}$$

$$= \frac{1}{18} + \frac{1}{9} + \frac{1}{9}$$

$$= \frac{1}{18} + \frac{4}{18}$$

$$= \frac{5}{18}$$

8. (1) $P(X_2=0 \mid X_0=0) = P_{00}^{(2)} = \frac{5}{9}$

$$P^2 = \begin{pmatrix} \frac{5}{9} & \frac{2}{9} & \frac{2}{9} \\ \frac{2}{9} & \frac{2}{9} & 0 \\ \frac{2}{3} & 0 & \frac{1}{3} \end{pmatrix}$$

$$P(X_0=0 \mid X_2=0) = \frac{P(X_0=0,\ X_2=0)}{P(X_2=0)} = \frac{\frac{5}{9}\times\frac{1}{3}}{\frac{5}{9}\times\frac{1}{3} + \frac{1}{3}\times\frac{2}{9} + \frac{1}{3}\times\frac{2}{3}} = \frac{\frac{5}{27}}{\frac{13}{27}}$$

$$= \frac{5}{13}$$

(2) $P(X_1=0) = \frac{1}{3}P_{00} + \frac{1}{3}P_{10} + \frac{1}{3}P_{20}$

$$= \frac{1}{3}\times\frac{1}{3} + \frac{1}{3}\times\frac{2}{3} + \frac{1}{3}\times 0$$

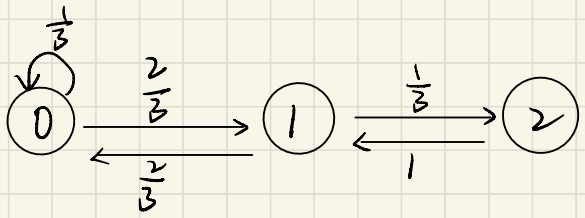
$$= \frac{1}{3}$$

$$P(X_1=0,\ X_3=0,\ X_4=1,\ X_6=1) = P(X_1=0)\cdot P_{00}^{(2)}\cdot P_{01}^{(1)}\cdot P_{11}^{(2)}$$

$$= \frac{1}{3}\times\frac{5}{9}\times\frac{2}{3}\times\frac{7}{9}$$

$$= \frac{70}{729}$$

(3)



$$f_{11}^{(n)} = \begin{cases} 0, & n=1 \\ \frac{7}{9}, & n=2 \\ \frac{4}{9}\cdot\left(\frac{1}{3}\right)^{n-2}, & n\geq 3 \end{cases}$$

$$f_{11} = \frac{7}{9} + \sum_{n=3}^{\infty} \frac{4}{9}\left(\frac{1}{3}\right)^{n-2}$$

$$= \frac{7}{9} + \frac{4}{9}\cdot\frac{\frac{1}{3}\left[1-\left(\frac{1}{3}\right)^{n}\right]}{1-\frac{1}{3}}$$

$$= \frac{7}{9} + \frac{4}{9}\left[1-\left(\frac{1}{3}\right)^{n}\right], \quad n\to\infty$$

$$= 1$$

$$M_1 = 2\times\frac{7}{9} + \sum_{n=3}^{\infty}\frac{4}{9}\cdot\left(\frac{1}{3}\right)^{n-2}\cdot n$$

$$S_n = \sum_{n=3}^{\infty}\left(\frac{1}{3}\right)^{n-2}\cdot n = 3\cdot\frac{1}{3} + 4\cdot\left(\frac{1}{3}\right)^{2} + \cdots\cdots + n\cdot\left(\frac{1}{3}\right)^{n-2}$$

$$\frac{1}{3}S_n = 0 + 3\cdot\left(\frac{1}{3}\right)^{2} + \cdots\cdots (n-1)\left(\frac{1}{3}\right)^{n-2} + n\cdot\left(\frac{1}{3}\right)^{n-1}$$

$$\frac{2}{3}S_n = 1 + \left(\frac{1}{3}\right)^{2} + \cdots\left(\frac{1}{3}\right)^{n-2} - n\cdot\left(\frac{1}{3}\right)^{n-1}$$

$$= 1 + \frac{\frac{1}{9}\left[1-\left(\frac{1}{3}\right)^{n-3}\right]}{1-\frac{1}{3}} - n\cdot\left(\frac{1}{3}\right)^{n-1}$$

$$= 1 + \frac{1}{6} - \frac{1}{6}\left(\frac{1}{3}\right)^{n-3} - n\cdot\left(\frac{1}{3}\right)^{n-1}$$

$$= \frac{7}{6} - \frac{1}{2}\cdot\left(\frac{1}{3}\right)^{n-2} - n\cdot\left(\frac{1}{3}\right)^{n-1}$$

$$= \frac{7}{6} - \left(\frac{3}{2}+n\right)\left(\frac{1}{3}\right)^{n-1}$$

$$S_n = \frac{7}{4} - \left(\frac{9}{4}+\frac{3}{2}n\right)\times\left(\frac{1}{3}\right)^{n-1}$$

$$M_1 = \frac{14}{9} + \frac{4}{9}\times\left[\frac{7}{4} - \left(\frac{9}{4}+\frac{3}{2}n\right)\times\left(\frac{1}{3}\right)^{n-1}\right]$$

$$= \frac{14}{9} + \frac{7}{9} - \left(1+\frac{2}{3}n\right)\times\left(\frac{1}{3}\right)^{n-1} \quad n\to\infty$$

$$M_1 = \frac{21}{9} = \frac{7}{3}$$